

AMAL BAJPAYEE

☎ +91-8318753734 ✉ amalbajpayee09@gmail.com [in LinkedIn](#) [G GitHub](#) [LeetCode](#)

Education

Atal Bihari Vajpayee Indian Institute of Information Technology and Management, Gwalior 2022-2027
Integrated B.Tech + M.Tech in Information Technology CGPA: 7.05
Relevant Coursework: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, Object Oriented Programming

Technical Skills

Languages: C++, Python, JavaScript, SQL
Backend: Node.js, Express.js, FastAPI
Frontend: React.js, HTML, CSS, TypeScript
Databases: MySQL, MongoDB, PostgreSQL
Tools: Git, GitHub, Linux, Postman, VS Code
Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Experience

Lord-Tech Datus Solutions Pvt. Ltd. Gwalior, India
Software Developer Intern 05/2026 – Present

- Contributing to the production-scale development of an AI-powered Smart City waste management platform using React, Node.js, FastAPI and PostgreSQL.
- Developing microservices, REST APIs and computer vision pipelines integrating YOLOv8, MiDaS and OpenCV, while collaborating on AWS-based cloud deployment.

Web Developer Intern 07/2025 – 12/2025

- Developed and customized client-facing web applications and dashboards based on project requirements.
- Debugged backend services and resolved API integration issues to improve application reliability and performance.
- Assisted in maintaining existing websites and implementing responsive user interfaces using modern web technologies.

Projects

NirmalGwalior — AI-Powered Smart City C&D Waste Management Platform [Web](#) — [App](#)
INR 2,00,000 TIIC Funded Project — Core Technical Member

- Secured **INR 2,00,000** innovation funding under TIIC, ABV-IIITM Gwalior for developing an AI-enabled civic-tech platform for construction and demolition waste management.
- Built a full-stack ecosystem comprising a citizen mobile application and a municipal dashboard for real-time waste reporting, monitoring and analytics.
- Developed computer vision pipelines using **YOLOv8, MiDaS and OpenCV** for waste classification, depth estimation and volume calculation from uploaded images.
- Designed a microservice-based architecture using **React, Node.js, FastAPI, PostgreSQL and Redis**, integrating GIS analytics and intelligent route optimization for scalable Smart City deployment.

Layer Sequence Extraction of Optimized DNNs Using Side-Channel Information Leaks 05/2025 – 07/2025
Bachelor Thesis Project — Python, PyTorch, Deep Learning, LSTM, Multi-Head Attention

- Investigated side-channel information leakage techniques for architecture inference of optimized deep neural networks.
- Explored sequence modeling approaches using recurrent neural networks and attention mechanisms for reconstructing DNN layer sequences.
- Studied execution-level GPU trace patterns and operator fusion mechanisms to understand their impact on model extraction and architecture reconstruction.

Achievements

- Secured **INR 2,00,000** innovation funding under TIIC, ABV-IIITM Gwalior for the NirmalGwalior Smart City project.
- Served as a **Core Technical Member** and Co-Principal Investigator for the AI-enabled NirmalGwalior platform.
- Successfully completed the **Ethical Hacking and Penetration Testing** program and virtual internship conducted by **C-DAC Noida** under the Cyber Gyan Project.